



الدراسة التحليلية والتصميمية لنظام إدارة المعلومات الصحية

Health Information Management System

مشروع القسم العملي من مادة هندسة البرمجيات 1
للسنة الرابعة

إعداد الطلاب:

بشار كميل بلان
صخر سلمان حاطوم
جنى نجيب جمول
قصي وسيم أبو خير
همسة بسام صوان نعيم

إشراف:

م. ماهر صارم

Software requirement specification

For

<Health Information Management System>

Version 1.0 approved

Prepared by Teen Titans

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1. Introduction

1.1 Purpose

This document outlines the requirements for a Health Information Management System aimed at automating and improving the management of health information in hospitals affiliated with the Ministry of health and ambulance services. The systems aims to provide an integrated environment for storing and managing patients medical data, which contributes to enhancing the quality of healthcare and the speed of response to emergency cases.

1.2 Product Scope

The Health Information Management System (HIMS) is designed to digitize and streamline hospital operations and emergency response systems for the Ministry of Health. The system will provide centralized management of patient unified medical records because it aims to improve decision-making for healthcare providers, enhance patient care quality, and reduce operational inefficiencies. It will also ensure quick response times for emergency services through automated ambulance dispatch and GPS tracking. and integration with laboratory, radiology, and pharmacy systems.

The product will:

- Maintain a unified medical record for each patient across all hospitals.
- Allow healthcare professionals to access patient information securely and in real time.
- Tracking the patient's condition from admission to discharge .
- Enable patients to request ambulance services and schedule appointments via a mobile application.
- Use GPS technology for efficient ambulance dispatch and tracking.
- Managing all hospital departments and enhancing communication between them to facilitate information exchange.

2.Overall Description

2.1 Product Perspective

The **Health Information Management System (HIMS)** is a new, self-contained product designed to modernize and automate the operations of hospitals and ambulance services under the Ministry of Health. This system is intended to serve as a centralized solution for managing patient medical records, hospital workflows, and emergency services.

Context and Origin:

- **Context:**

The HIMS is being developed as part of the Ministry of Health's digital transformation initiative to improve healthcare delivery and emergency response systems. It replaces outdated manual processes and legacy systems used for patient record management, appointment scheduling, and ambulance dispatching.

- **Origin:**

The product is built from the ground up, incorporating modern technologies to ensure scalability, security, and user-friendliness. It is not part of an existing product family but will integrate with existing hospital databases and third-party services where applicable.

2.2 Product Functions

The Health Information Management System (HIMS) will support the following primary functions:

Management of Doctors' Accounts

- Login and logout records
- Adding, editing, and deleting information and appointments
- Alerts for surgery dates

Department Management

- Communication and coordination between departments for information exchange
- Managing staff in the department, including doctors and nurses
- Managing resources associated with the department

Patient Records Management

- Creating and updating medical records
- Displaying the unified medical record based on the national ID number.

Appointment Scheduling

- Allow patients to book and cancel appointments online or through hospital staff.

- Enable doctors to manage their schedules and view upcoming appointments

Emergency Services and Ambulance Management

- Allow patients to request ambulances via a mobile app or hotline.
- Dispatch ambulances automatically based on location, availability, and priority.

Diagnostic and Treatment Management

- Manage orders for laboratory tests, imaging, and other diagnostic services.
- Store and share test results with doctors and patients.
- Record treatment plans and monitor progress.

Billing

- Generate bills for services provided to patients.
- Allow patients to make payments through integrated gateways.

Reporting and Analytics

- Generate reports on hospital performance, patient demographics, and operational metrics.
- Track the availability of medicines, medical equipment, and hospital beds.
- Alert staff about low inventory levels to ensure timely restocking.

User Access and Security

- Provide role-based access control to ensure data security.
- Maintain audit logs of all user activities for accountability.

. System Administration

- Allow administrators to configure system settings, user roles, and permissions.

- Monitor system performance and manage backups.

2.3 User Classes and Characteristics

Primary Actors

These are the main users who interact with the system to perform specific tasks.

1. Ministry of Health

- Access aggregated data for policymaking and public health monitoring.
- Hospital management within the system in terms of addition and deletion.

2. Hospital Management/Administrators

- Monitor overall hospital operations.
- Generate reports on patient care, staff performance, and resource utilization.

3. Reception staff

- Handling patient check-in procedures.
- Managing discharge procedures and directing patients to the accounting departments.

4. Department Head

- Manage the department's information, including equipment, medications, and materials related to the department
- Manage the doctors and nurses affiliated with the department
- Manage the rooms associated with the department and how to distribute patients
- Prepare reports
- Prepare and coordinate doctors' schedules and facilitate coordination among them

5. Patients

- Book an appointment at the hospital through the app.
- Request ambulance services in emergency situations.
- Access their medical records.
- Receive reminders for examination or surgery appointments.
- Receive test results

6. Doctors

- Manage their personal account.
- Modify their appointments and procedures
- Access and update patient medical records.
- Communicate with other departments to exchange medical documents(e. , lab test, imaging).
- Develop, monitor, and update treatment plans

7. Nurses

- Manage patient records and care plans.
- Monitor patient progress in real time.

8. The department secretary:

- Coordinates patient admission appointment.
- Approval of reservations based on the doctor's duty schedule

9. Ambulance Staff

- Receive and prioritize emergency requests.
- Assign ambulances based on location and availability.
- Monitor ambulance locations via GPS in real time.
- Receive navigation instructions and patient details.

- Communicate with dispatchers for updates on assignments.
- Provide updates about their status (route, arrived, completed).

10. Pharmacists

- Manage inventory of medicines.
- Create report about medicines information.

11. Lab Technicians

- Receive test orders from doctors.
- Update test results in the system.
- Send the results to the concerned department.

12. Timer

- Sending notifications to alert doctors and patients before a certain period of time (24 hour) for examinations or operations.

13. Locater

- Manage alerts regarding departure, arrival and route information.

Secondary Actors

These actors do not directly interact with the system but influence or rely on its functionality.

1. Third-Party Services

- **GPS Services:** Provide location data for ambulance dispatch.

- **Payment Gateways:** Facilitate payment processing for hospital services.

2. IT Support Staff

- Maintain and troubleshoot the system.
- Ensure data security and uptime.

2.4 Constraints

1. Technical Constraints

- The system must be web-based and support modern browsers (e.g., Chrome, Firefox, Edge).
- Mobile applications must support Android (version 8.0 and above) and iOS (version 12 and above).

2. Regulatory Constraints

- The system must comply with the Health Insurance Portability and Accountability Act (HIPAA) to ensure the secure handling and confidentiality of patient data.
- The system must adhere to applicable local healthcare regulations and data protection laws, including [insert specific regulations if known, e.g., GDPR, CCPA]

3. Performance Constraints

- The ambulance dispatch system must process requests and determine the nearest available ambulance within 5 seconds

- The system must support simultaneous access by up to 1,000 users without degradation in performance.

4. Environmental Constraints

- The system must function reliably in areas with limited internet connectivity
- Offline access should be provided for specific modules (e.g., ambulance tracking).

5. Interoperability Constraints

- The system must integrate with existing national health databases .
- Support for standardized data formats (e.g., HL7 or FHIR for healthcare data).

6. Design Constraints

- The interface must be responsive, ensuring usability on both desktop and mobile devices
- The system must support at least three languages (English and Arabic).

7. Hardware Constraints

- The system must operate on servers with the following minimum specifications:
 - o Processor: Intel Xeon 2.5 GHz or equivalent.
 - o RAM: 16 GB.
 - o Storage: 1 TB.
- Mobile applications must work on devices with at least 4 GB of RAM.

2.5 Assumptions and Dependencies

Assumptions

1. Infrastructure Availability:

- Hospitals and ambulance stations will have reliable internet connectivity for system operations.

2 . User Readiness:

- Staff and patients will have basic familiarity with computers or smartphones

3. Timely Resource Allocation:

- Sufficient funding and human resources will be available throughout the development lifecycle.

4. Hardware Compatibility:

- The system will be deployed on servers and devices that meet the specified hardware requirements.

Dependencies

1. Third-Party Services:

- The system relies on GPS services for real-time ambulance tracking.

- Integration with national healthcare databases for retrieving patient histories.

2. External Software:

- Use of third-party libraries or APIs for data visualization and reporting.

3. Regulatory Compliance:

- Adherence to healthcare data standards like HL7 or FHIR.

4. Organizational Policies:

- The success of the system depends on hospital staff adhering to updated workflows and processes.

5. Technology Stack:

- The system assumes continued support for the chosen programming languages, frameworks, and databases

3.System Feature

3.1 Hospital Information Management:

The system allows the Ministry of health to manage the hospital and the hospital manager to access hospital information through a web dashboard.

Priority: Medium

Req.1- The system must allow the ministry to add new hospital.

Req.2- The system must allow the ministry to revoke hospital.

Rqe.3- The system must allow the manager to view, add, delete, and modify employee information.

Rqe.4- The system must allow the manager to view the hospital departments and their contents, as well as make modifications.

3.2 Department Information Management:

The system enables the department head, who is a physician, to view and manage department information through a web interface.

Priority: High

Rqe .1- The system must allow the department head to view the staff in this department, including doctors and nurses, and make modifications.

Rqe .2- The system must allow the department head to display, add, delete, and modify materials and supplies available in the department.

Rqe .3- The system must allow the department head to send requests to the relevant departments for materials and equipment when needed.

Rqe .4- The system must allow the department head to manage patient rooms associated with the department, display their information and numbers, and coordinate between patient cases and rooms.

3.3 Scheduling and Coordination of Appointments

The system enables the management of appointments and coordination between patients and doctors, as well as among doctors themselves.

priority : medium.

Rqe .1- The system allows the head of the department to create duty schedules that coordinate the work of doctors with each other, in addition to modifying and deleting these schedules in coordination with the doctors.

Rqe .2- The system allows doctors to input the times they are available so that duty schedules can be coordinated accordingly.

Rqe .3- The system permits doctors to cancel appointments within a specified period before the actual appointment time.

Rqe .4- The system allows both doctors and patients to view doctors' appointment schedules.

Rqe .5- The system enables mobile app users (patients) to book appointments based on what is available according to the appointment schedules.

Req.6 - The system must allow the department secretary to view the appointment schedule reserved by patients.

3.4 Patient Records Management

The system provides a unified patient record that includes all medical information and the patient's medical history.

Priority: High

Rqe .1- The system should allow for the display of patient records based on the national ID for doctors, nurses, and reception staff through the web application, and for patients through the mobile application.

Rqe .2- The system should enable doctors to add new records and modify existing records based on patient information.

3.5 Patient Status Monitoring

The system enables tracking of a patient's condition from their admission to the hospital until their discharge.

Priority: High

Rqe .1- The system must allow reception staff to register the patient's admission to the hospital and, upon completion of treatment, to carry out discharge procedures.

Rqe.2- The system must allow doctors to create treatment plans, modify them, and review them.

Rqe .3- The system must allow nurses to access treatment plans.

3.6 Notification and Alert System

The system allows for sending notifications to users.

Priority: medium.

Rqe .1- The system must enable the timer to send alerts to both doctors and patients to remind them of surgery appointments 24 hours before the actual appointment time.

Rqe .2- The system must allow the timer to send a notification to the patient when the doctor cancels an appointment, provided that the cancellation occurs 24 hours or more before the actual appointment time.

3.7 Reports and Statistics

The system enables the generation of detailed reports and statistics.

Priority: medium.

Rqe .1- The system must allow the hospital manager to send reports to the Ministry of Health.

Rqe .2- The system must allow web application users to send and receive reports related to either patients or departments.

3.8 Ambulance Request System

A system that allows users to easily request an ambulance through a mobile application.

Priority: High

Rqe.1- The system must allow patients to request an ambulance while entering their location.

Rqe.2- The system must allow tracking of the ambulance's route and the estimated time of arrival.

3.9 Vehicle Location:

Utilizing the Global Positioning System (GPS) to determine the location of ambulances.

Priority: High

Rqe.1- The system must allow the use of GPS to ensure the ambulance reaches the desired location in the shortest possible time.

3.10 Medication and Inventory Management

The system allows for the management of medications and medical supplies within the hospital.

Priority: Medium

Rqe.1- The system must allow the warehouse manager to enter materials, modify their quantities, and delete them.

Rqe.2 - The system must allow the warehouse manager to respond to the department head's requests, if possible, by sending a report that includes how the request was addressed and the quantities involved.

3.11 Account Management

The system allows users to manage their accounts.

Priority: Medium

Req.1- The system must allow the hospital manager to create accounts for his employees with specified access permissions.

Req.2- The system must allow its users, including doctors, nurses, and patients, to log in to their accounts and make modifications.

Req.3-The system must allow mobile application users to create an account.

3.12 Integration with Other Applications

The system enables interaction with other applications.

Priority: Medium

Req.1- The system must allow patients to use electronic payment features through payment gateways.

Req.2-The system must allow patients to use GPS to determine their locations when requesting ambulances.

4.Non-Functional Requirements

4.1 Performance Requirements

NReq.1-The system shall handle at least 1,000 concurrent users without performance degradation.

NReq.2-The system shall respond to user actions (e.g., record retrieval, appointment scheduling) within 2 seconds on average under normal load conditions.

NReq.3-The system shall process and display emergency requests, such as ambulance dispatch, within 5 seconds.

NReq.4-The system must process a minimum of 500 transactions per second (TPS) under normal load, with scalability up to 2,000 TPS during peak hours.

NReq.5-Data synchronization between the centralized database and remote hospital systems shall occur in less than **5 seconds** for all updates.

NReq.6- The system shall maintain real-time location tracking for ambulances with an update frequency of 1 second.

NReq.7- The system must scale to 10,000 concurrent users during emergencies or peak times by utilizing cloud resources or load balancing.

NReq.8- The system shall process large datasets, such as medical imaging results or batch updates to patient records, in under **10 seconds** per 100MB of data.

4.2 Safety Requirements

4.2.1 Patient Safety:

- **Accurate patient identification:**

NReq .1- The system must ensure correct patient identification using **unique identifiers** (e.g., ID numbers, barcodes) to prevent misdiagnosis, incorrect treatments, or medication errors.

NReq .2- Duplicate records must be flagged and resolved.

- **Role-Based Access Control:**

NReq .1- Limit system access based on user roles (e.g., doctor, nurse, admin) to prevent unauthorized modifications to critical medical data.

- **Error Prevention for Critical Data Entries:**

NReq .1- Implement data validation for entries such as **medications, dosages, allergies, and medical history** to prevent errors.

NReq .2- High-risk operations (e.g., discharging a patient or approving treatment plans) must require **double confirmation** or approval from senior staff.

4.2.2 System Safety and Reliability:

- **System Redundancy and Backup:**

NReq .1- The system must include redundant servers and regular backups to prevent data loss in the event of hardware failures, software crashes, or power outages.

NReq .2- Automatic failover must ensure continuous availability of critical functions such as emergency response and patient monitoring.

- **Graceful Degradation:**

NReq .1- In the event of a system failure, the system must allow basic operations like **ambulance dispatch, vital alerts, and patient record access** to continue in offline or degraded mode.

NReq .2- Full recovery of the system must occur within **30 minutes**.

- **Training and User Guidance:**

NReq .1- Medical staff and system users must receive training on system usage and patient safety protocols.

NReq.2- Provide on-screen instructions or warnings during critical actions to minimize errors.

4.2.3 Ambulance Dispatch and Emergency Response

- **Reliable Ambulance Tracking:**

NReq .1- The system must track ambulances in **real time** and provide accurate location data to ensure timely arrival at emergency locations.

NReq .2- In case of GPS failure, the system must allow manual intervention for location tracking and dispatch.

- **Emergency Route Optimization:**

NReq .1- The system must integrate with mapping tools to provide the **fastest route** for ambulances based on real-time traffic conditions.

- **Fail-Safe for Communication:**

NReq .1- Communication failures with ambulances must trigger **backup protocols**, such as sending location and dispatch details via SMS or alternate channels.

4.2.4 Safeguards Against Malicious Activity

- **System Monitoring for Anomalous**

NReq .1- The system must monitor and detect unusual activity (e.g., multiple failed login attempts, sudden bulk data downloads) and trigger **alerts**.

- **Unauthorized Access Prevention:**

NReq .1- The system must prevent unauthorized access to sensitive patient data and emergency features, such as ambulance dispatch.

4.3 Security Requirements

4.3.1. User Identity Authentication

- **Role-Based Access Control (RBAC):**

NReq .1- Users are assigned roles (e.g., doctor, nurse, admin) with corresponding permissions. Access to sensitive data and system functions must be restricted based on these roles.

- **Session Management:**

NReq.1- Enforce session timeouts after **15 minutes** of inactivity. Require secure reauthentication for resuming sessions.

- **Password Policy:**

NReq .1- Enforce a strong password policy, requiring passwords to be at least **8 characters long**, with a combination of letters, numbers, and special characters.

4.3.2. Data Protection

- **Encryption of Data at Rest:**

NReq .1- All stored data, including medical records and personal information, must be encrypted using **AES-256** or an equivalent standard.

- **Encryption of Data in Transit:**

NReq.1- Data exchanged between system components or with external entities must be encrypted using **TLS 1.3** or higher to prevent interception.

- **Secure Data Backup:**

NReq .1- Regular encrypted backups must be stored securely on both on-site and off-site locations to prevent data loss.

4.3.3 Data Privacy and Compliance

- **Data Minimization:**

NReq 1- Collect and process only the minimum amount of data necessary for medical and operational purposes.

4.4. Software Quality Attributes

NReq .1-Adaptability:

The system shall support customization for regional requirements, such as language settings and regulatory compliance.

NReq .2- Availability:

- The system shall maintain an uptime of **99.9%**, with failover mechanisms to ensure continuous availability.
- Failover machine must activate automatically within 10 seconds of detecting a primary server failure.

NReq.3- Correctness:

The system shall undergo rigorous testing to ensure accuracy in patient data, billing, and resource allocation.

NReq .4- Flexibility:

The system shall allow for the seamless addition of new modules, such as telemedicine or AI-based diagnostics.

NReq .5- Interoperability:

The system shall integrate with external systems like government databases and wearable health devices using **standard APIs** (e.g., HL7, FHIR).

NReq .6- Maintainability:

The system shall follow a modular architecture to facilitate maintenance and updates without affecting core functionality.

NReq .7- Portability:

The system shall run on major operating systems (Windows, Linux, macOS) and support major browsers (Chrome, Firefox, Safari, Edge).

NReq .8- Reliability:

The system shall ensure transactional integrity, even during high-load operations, to prevent data corruption or loss.

NReq .9- Reusability:

Common components, such as authentication or billing modules, shall be designed for reuse across other healthcare applications.

NReq .10- Robustness:

The system shall handle unexpected inputs or errors gracefully, displaying user-friendly error messages.

NReq .11-Testability:

The system shall provide a testing framework for automated and manual testing of all modules.

NReq .12- Usability:

The system's interface shall follow **best UX practices**, ensuring ease of navigation for both technical and non-technical users.

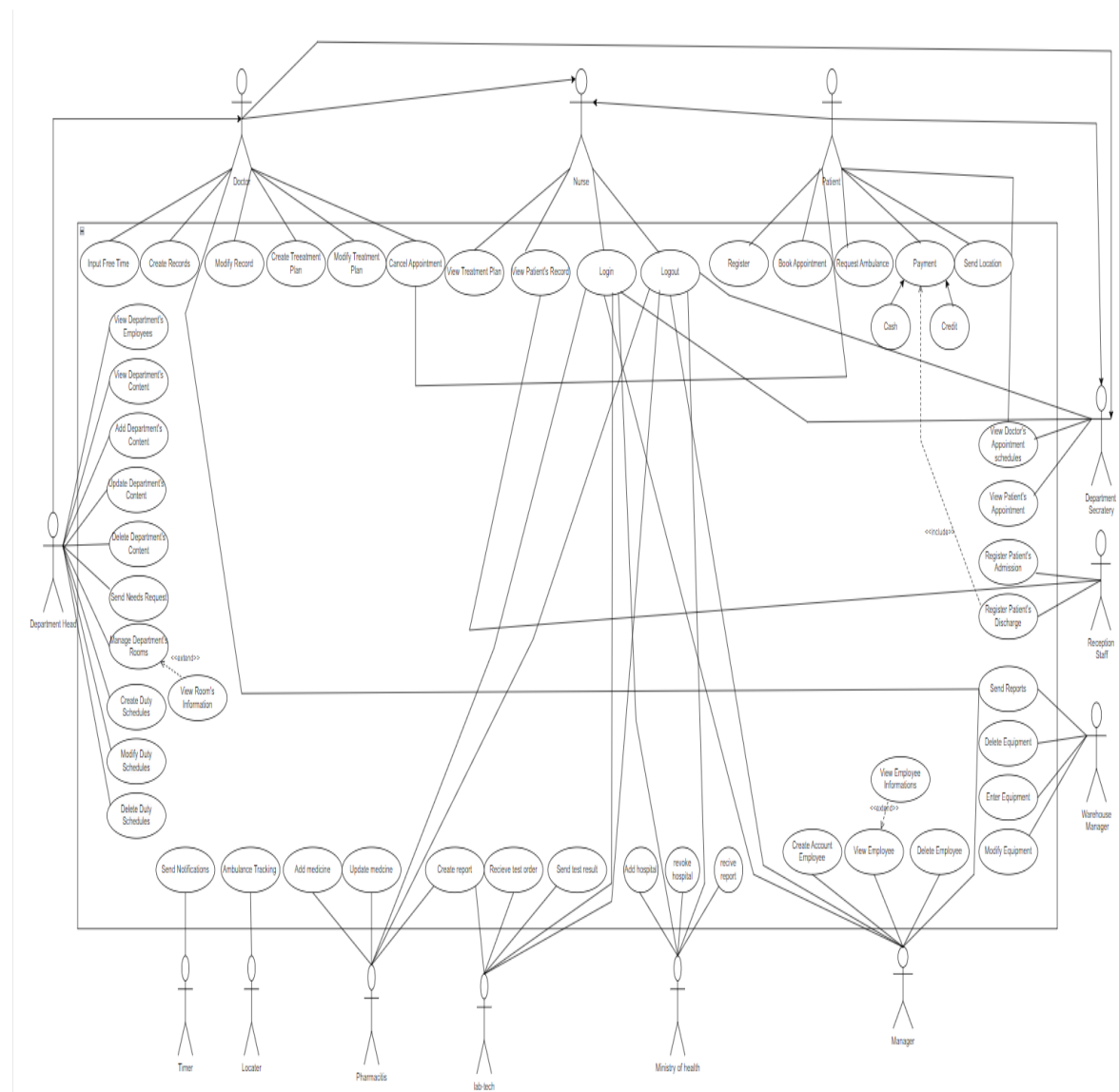
4.5 Business Rules

1. Only authorized medical staff can update patient medical records.
2. Patients or their guardians must provide explicit consent before sharing medical data with external entities.
3. Emergency ambulance requests can only be initiated by verified users or hospital staff.

4. Critical patient information, such as allergies and chronic diseases, must always be highlighted in patient profiles.
5. The system shall allow administrators to configure operating hours and schedules for hospital departments dynamically.
6. Access to financial data, such as billing records, is restricted to authorized finance personnel.
7. Automated ambulance dispatch must prioritize requests based on urgency levels (e.g., critical, non-critical).
8. Telemedicine consultations must comply with local telehealth regulations, including digital signature requirements for prescriptions.
9. The system must ensure that staff accessing sensitive data, such as mental health records, has specialized clearance.
10. Patients shall not be discharged until all pending bills are cleared, unless approved under an exception policy.

UML DIAGRAM

Use Case Diagram



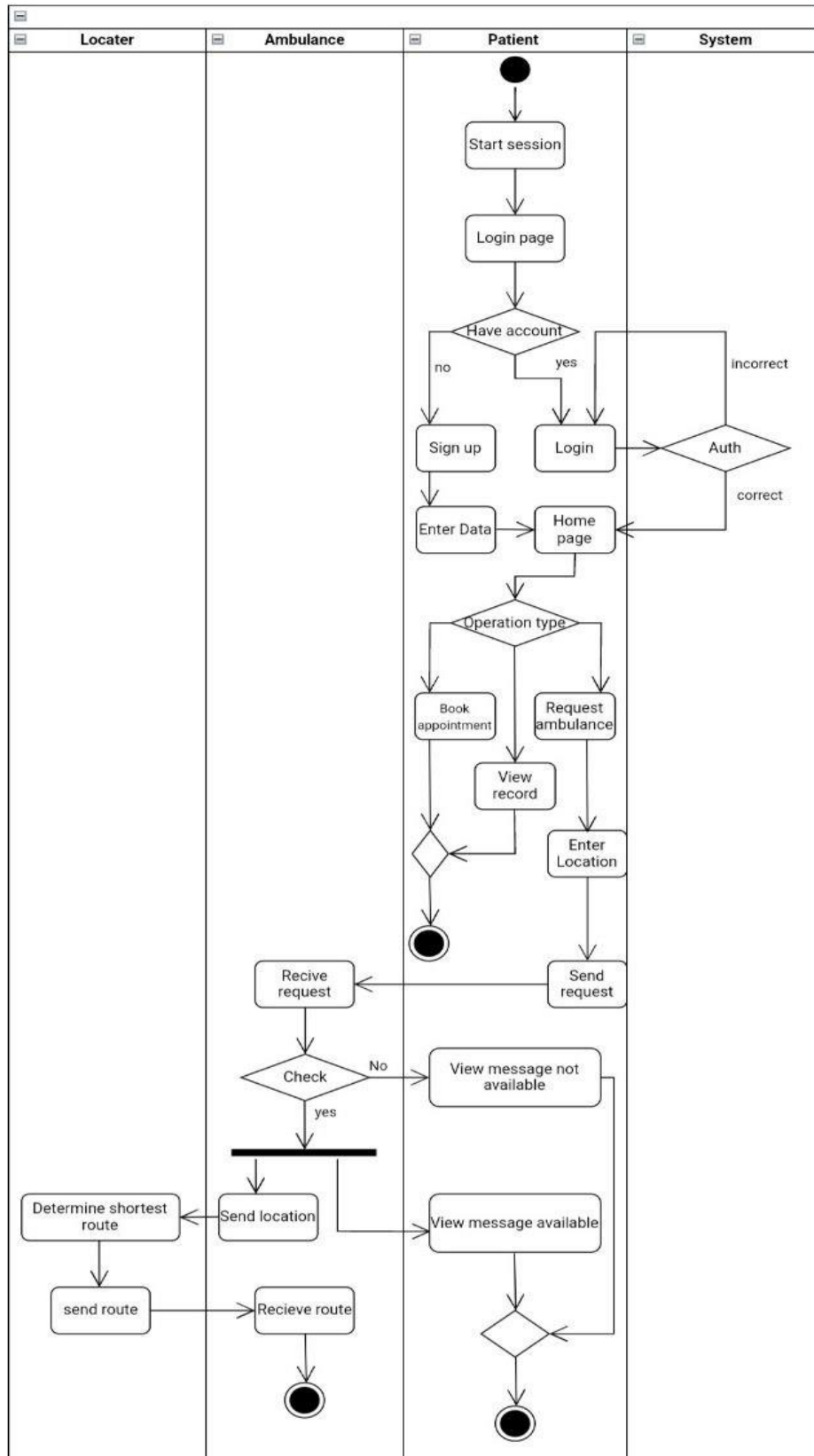
use case table

Use case ID	Actor	Use case
UC-1	Department Head / Doctor	Input free time
UC-2	Department Head / Doctor	Create records
UC-3	Department Head / Doctor	Modify record
UC-4	Department Head / Doctor	Create treatment plan
UC-5	Department Head / Doctor	Modify treatment plan
UC-6	Department Head / Doctor / Patient	Cancel Appointment
UC-7	Department Head / Doctor / Warehouse Manager / Manager	Send reports
UC-8	Department Head / Doctor / Patient / Nurse	View treatment plan
UC-9	Department Head / Doctor / Patient / Nurse / Reseption Staff	View patient's record
UC-10	Department Head / Doctor / Patient / Nurse / Department Secratery / Manager / Ministry of helth / Lab-tech / Pharmacitis	Login

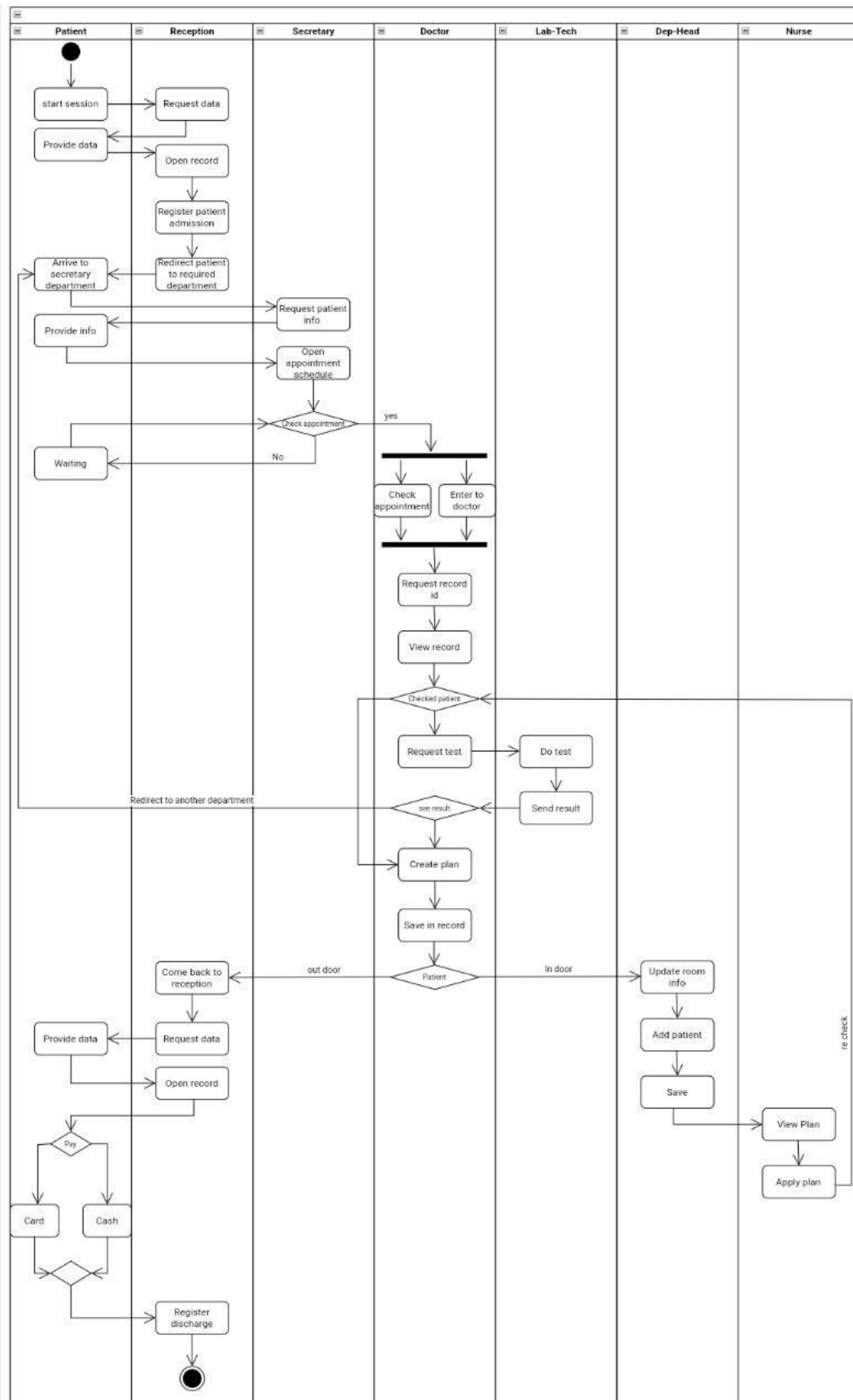
UC-11	Department Head / Doctor / Patient / Nurse / Department Secratery / Manager / Ministry of helth / Lab- tech / Pharmacitis	Logout
UC-12	Patient	Register
UC-13	Patient	Book appointment
UC-14	Patient	Request ambulance
UC-15	Patient	Payment
UC-16	Patient	Send location
UC-17	Department Head / Doctor / Patient / Department Secratery	View doctor's appointment schedules
UC-18	Department Head / Doctor / Patient / Department Secratery	View patient's appointment
UC-19	Reception Staff	Register patient's admission
UC-20	Reception Staff	Register patient's discharge
UC-21	Warehouse Manager	Delete equipments
UC-22	Warehouse Manager	Enter equipments
UC-23	Warehouse Manager	Modify equipments
UC-24	Manager	Delete employee
UC-25	Manager	View employee
UC-26	Manager	View employee information
UC-27	Manager	Create account employee
UC-28	Ministry of helth	Recive report
UC-29	Ministry of helth	Revoke hospital

UC-30	Ministry of helth	Add hospital
UC-31	Lab-tech	Send test result
UC-32	Lab-tech	Recive test order
UC-33	Lab-tech / Pharmacitis	Create report
UC-34	Pharmacitis	Update medicine
UC-35	Pharmacitis	Add medicine
UC-36	Locater	Ambulance tracking
UC-37	Timer	Send notifications
UC-38	Department Head	Delete duty schedules
UC-39	Department Head	Modify duty schedules
UC-40	Department Head	Create duty schedules
UC-41	Department Head	Manage department's rooms
UC-42	Department Head	View room's information
UC-43	Department Head	Send needs request
UC-44	Department Head	Delete department's content
UC-45	Department Head	Update department's content
UC-46	Department Head	Add department's content
UC-47	Department Head	View department's content
UC-48	Department Head	View department's employees

Activity Diagram For Request Ambulance

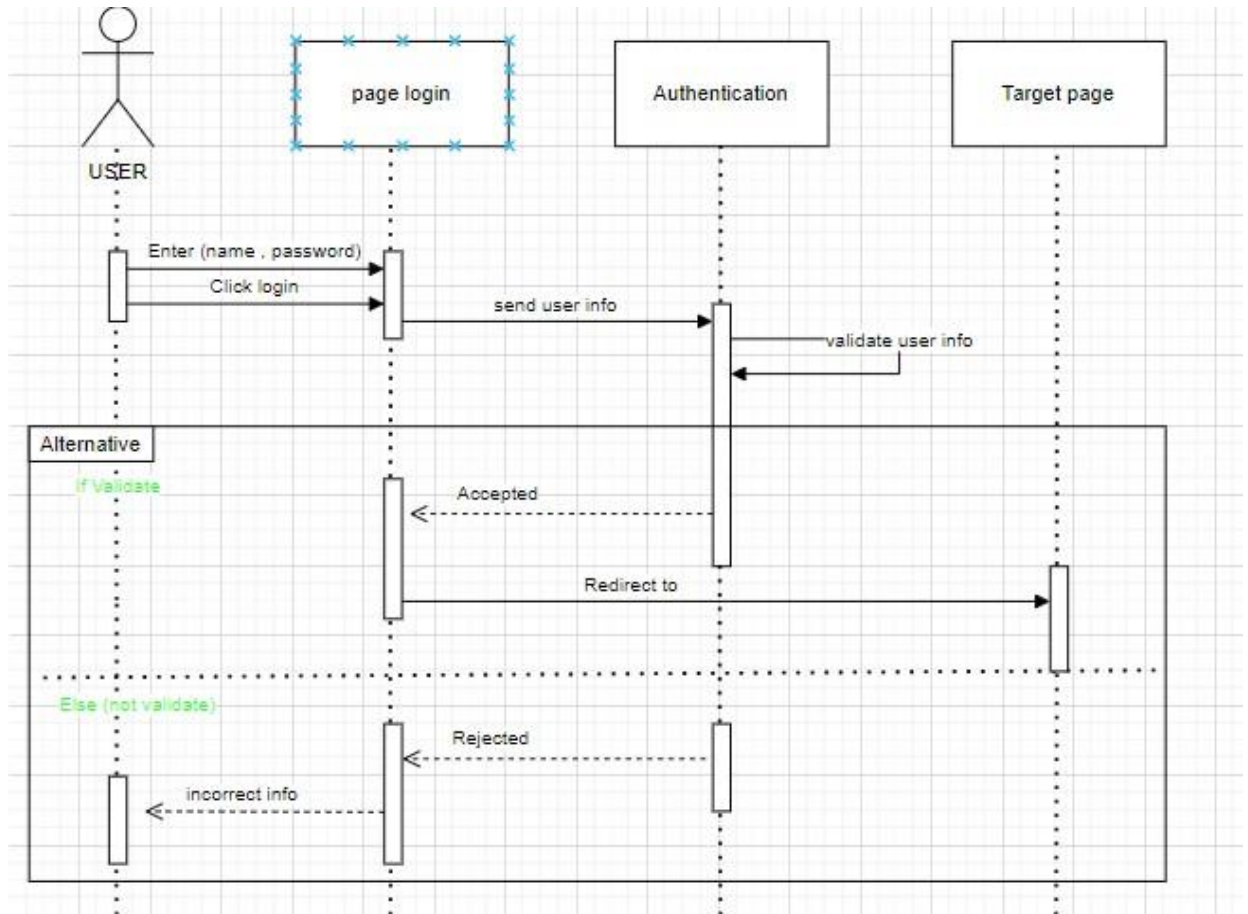


Activity Diagram For Register Patient

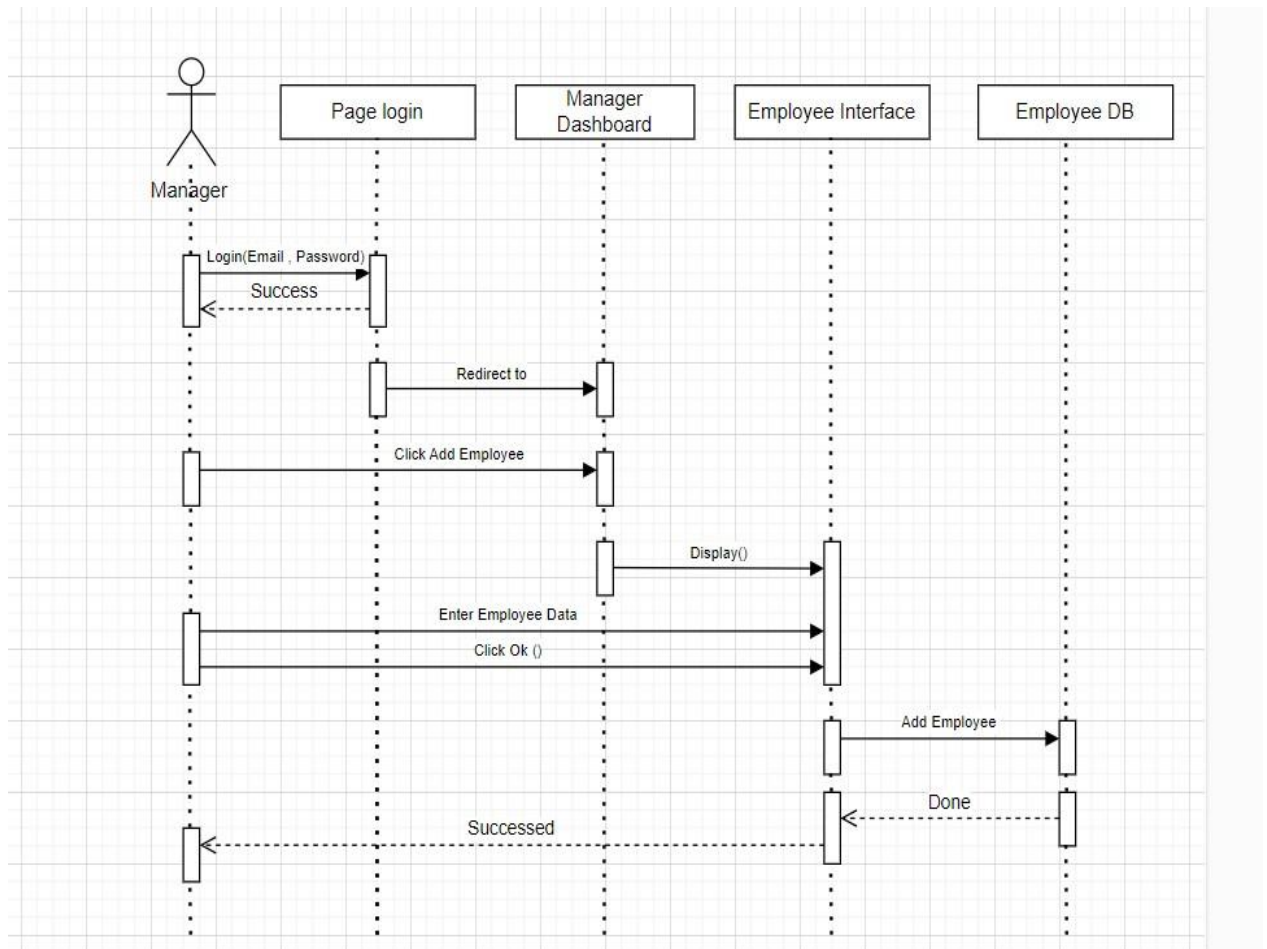


Sequence Diagram For:

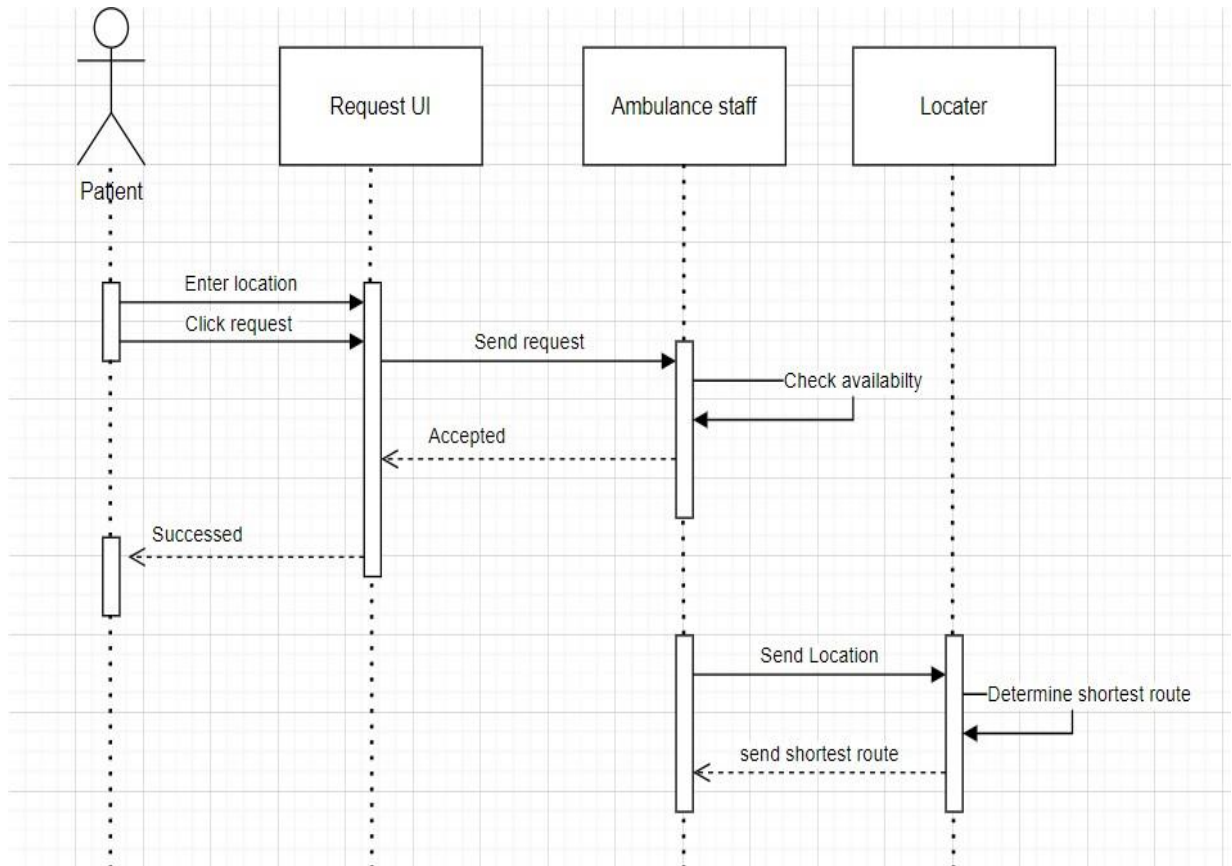
1. Log In:



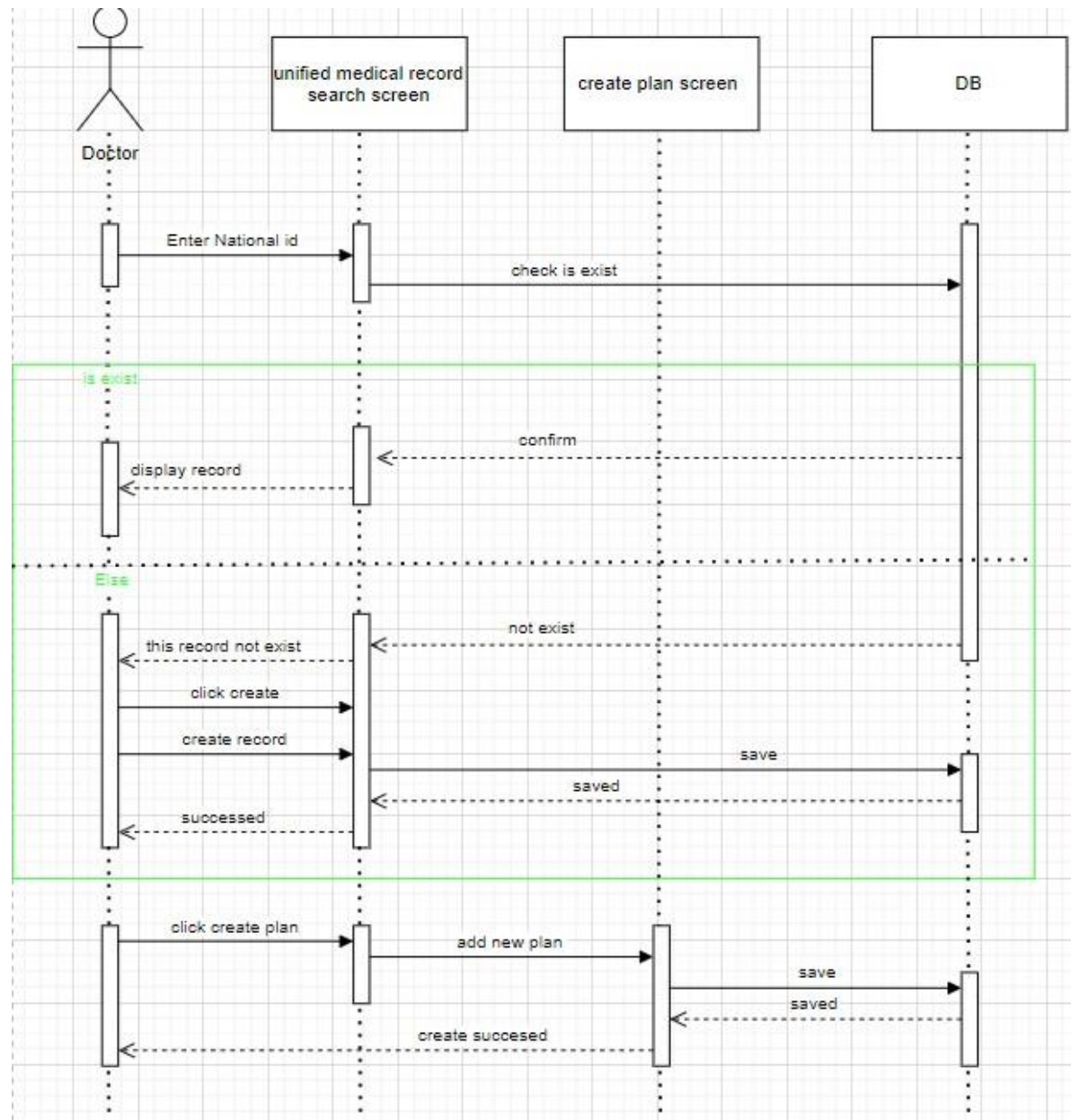
2. Add Employee:



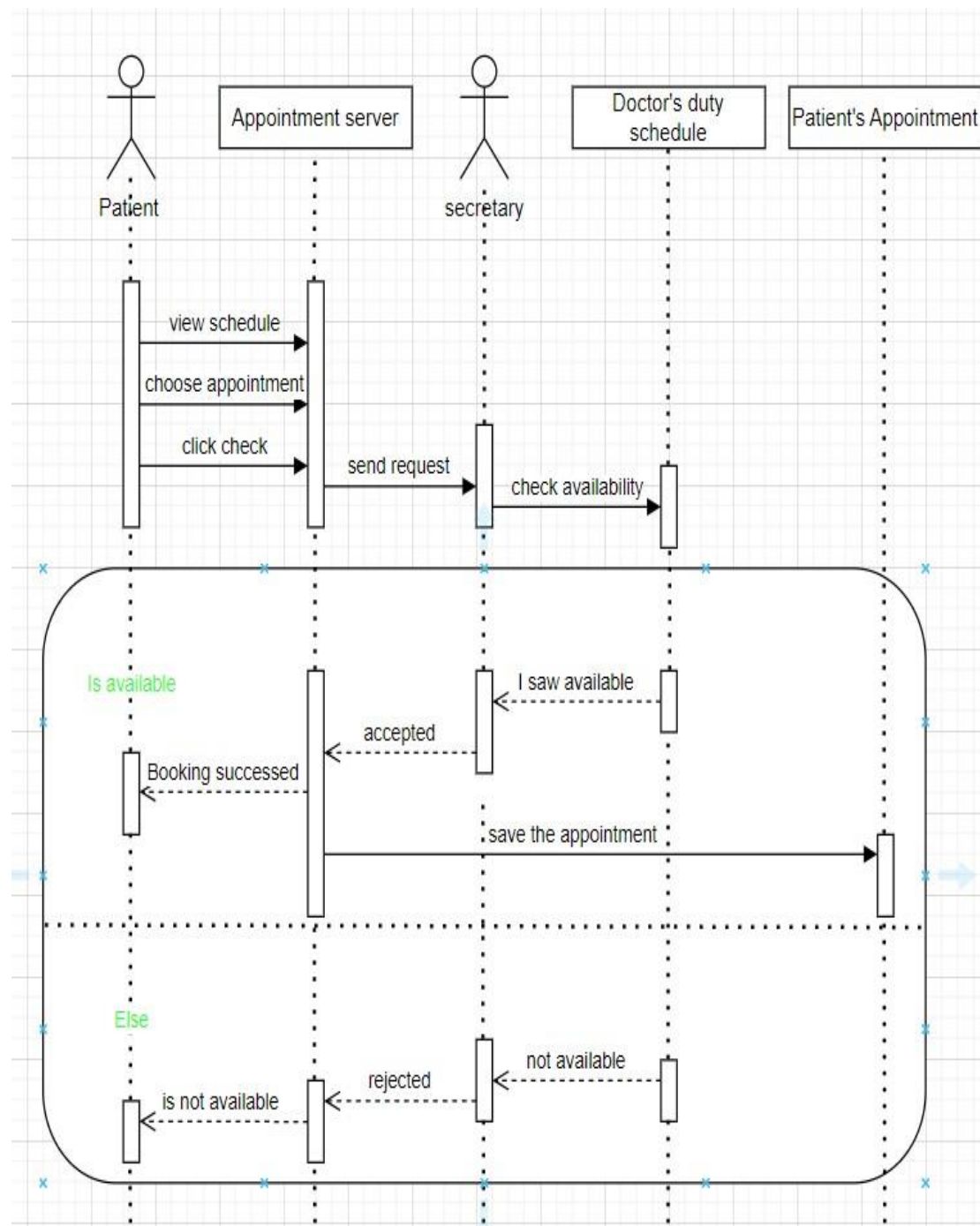
3. Request Ambulance:



4. Create Treatment Plan:

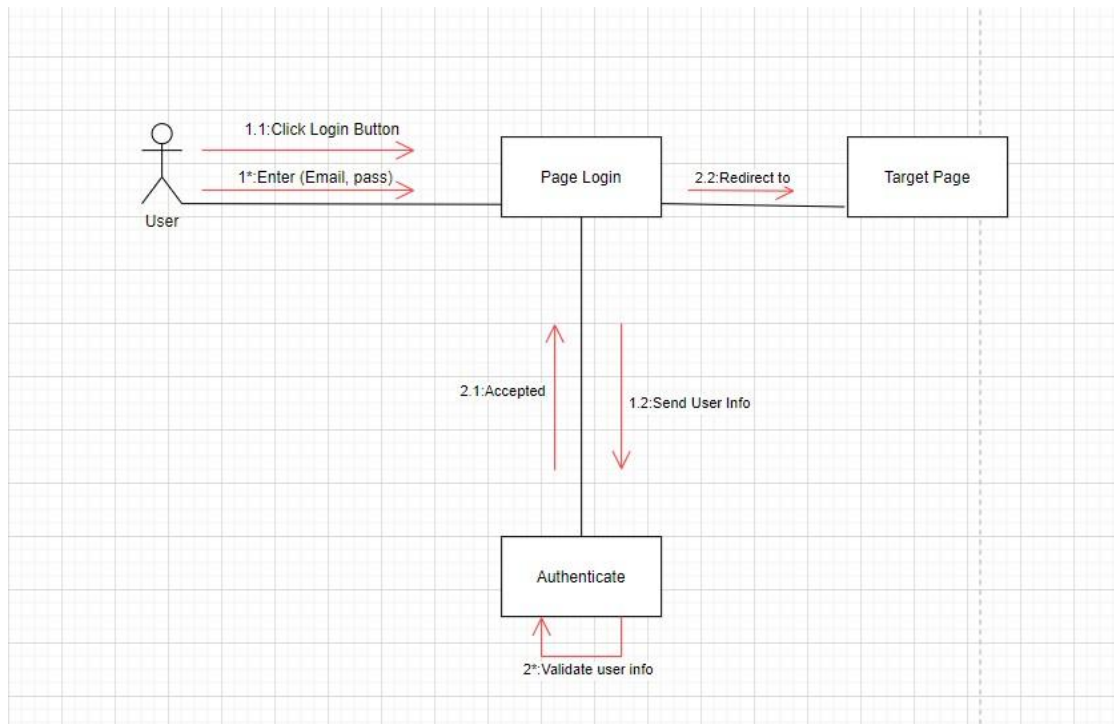


5. Book Appointment

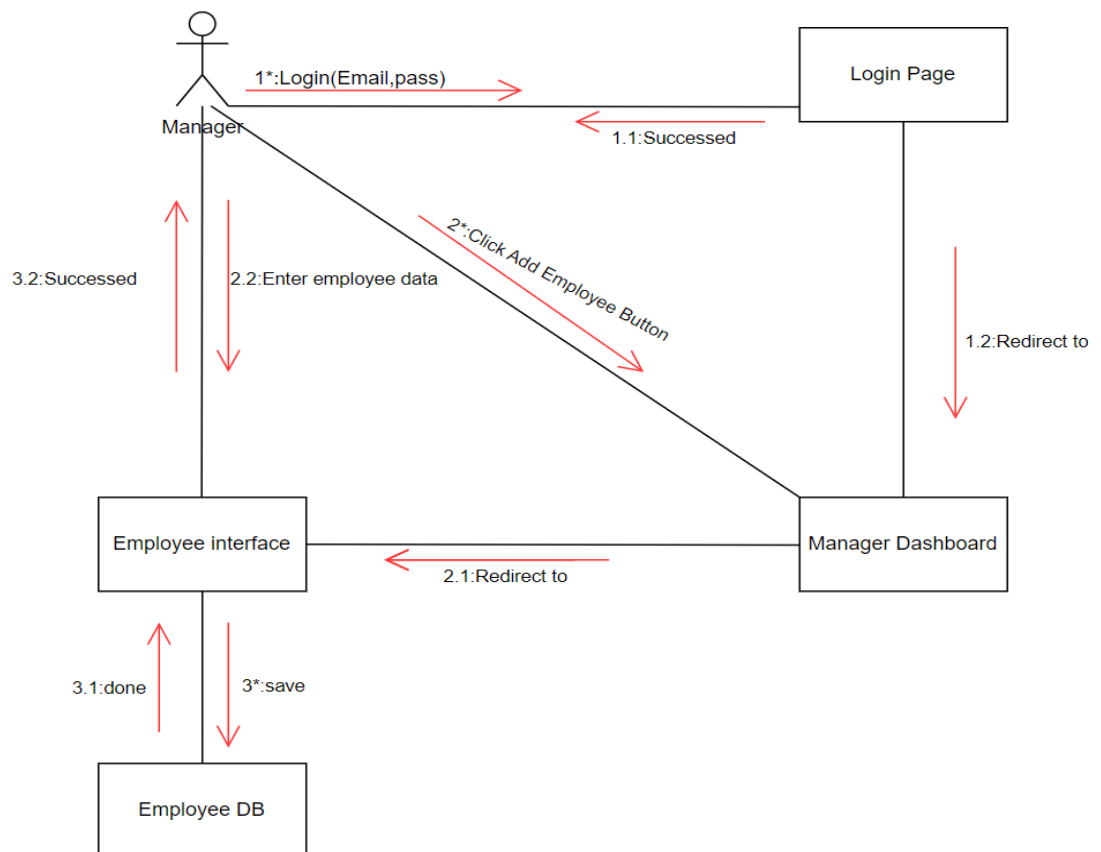


Collaboration Diagram For:

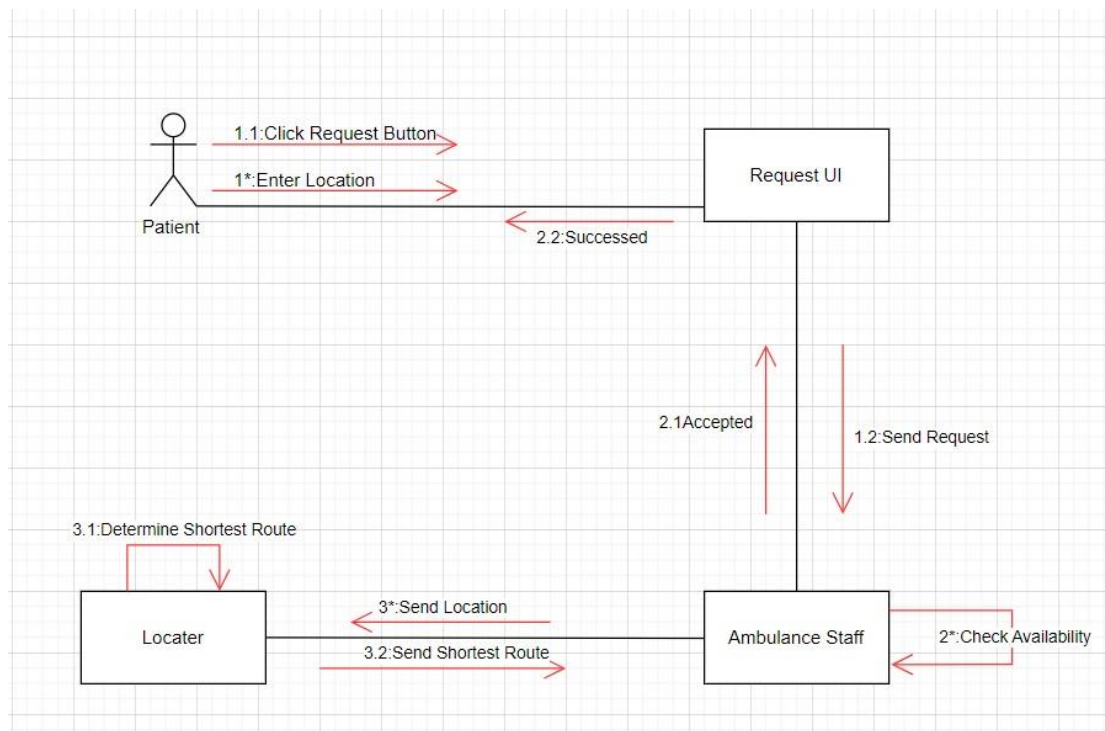
1. Log In



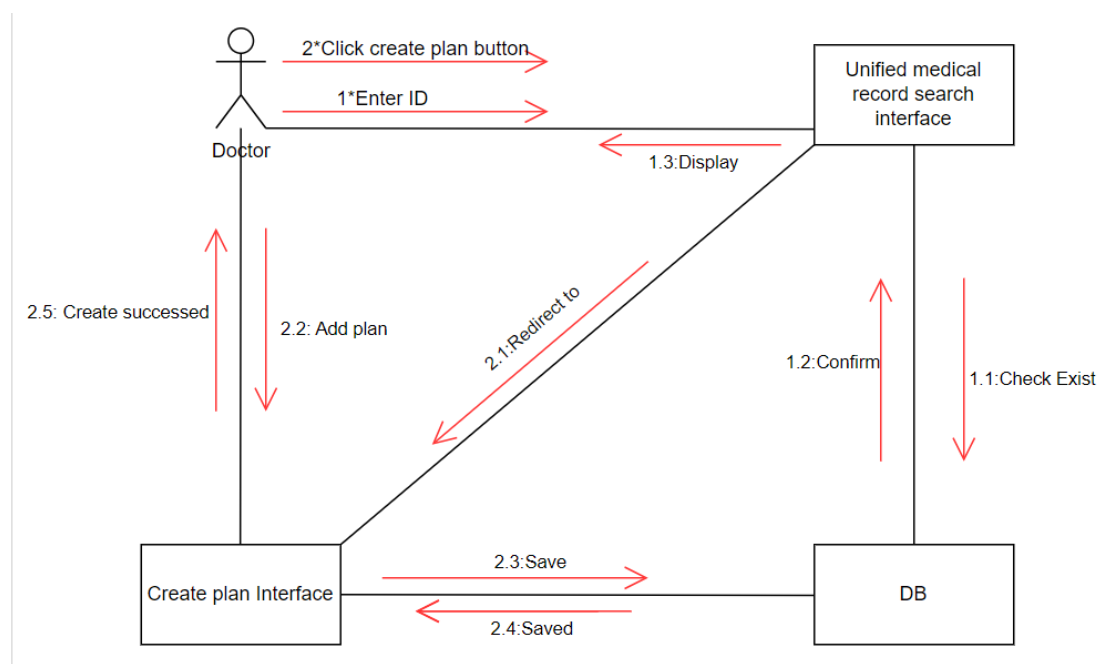
2. Add Employee:



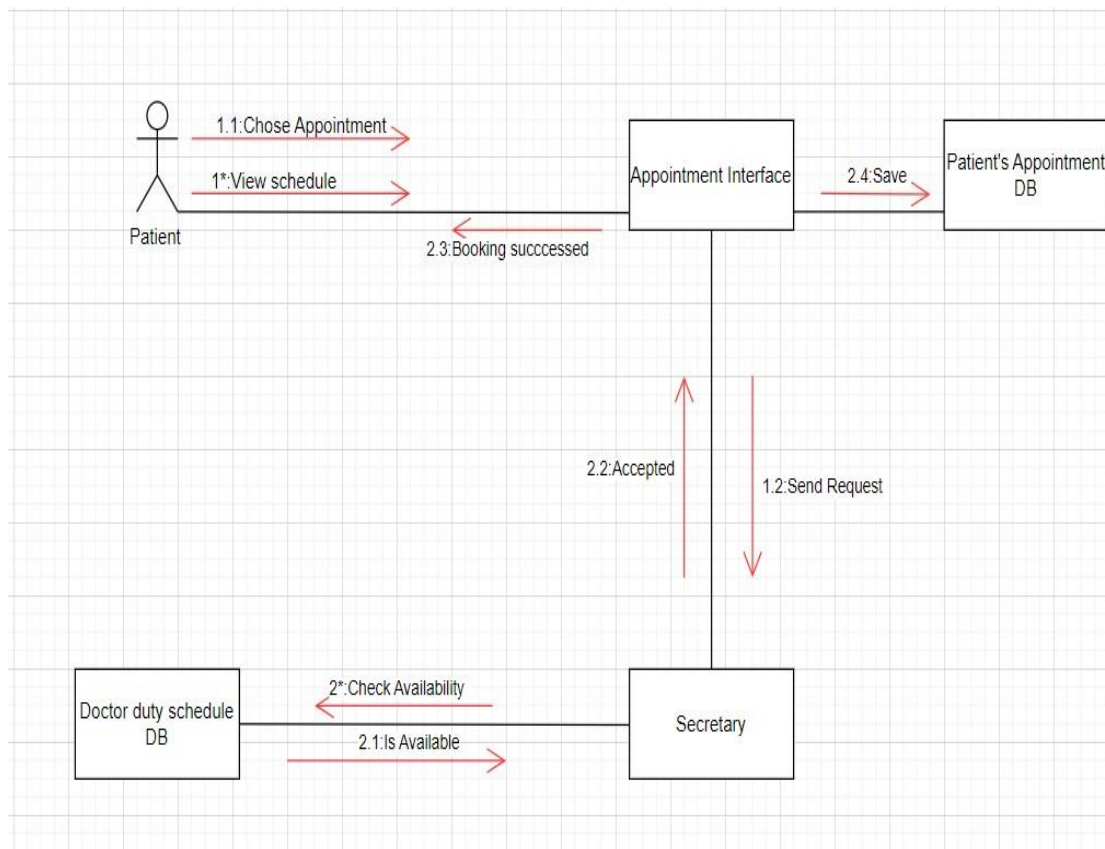
3. Request Ambulance:



4. Create Treatment Plan:



5. Book Appointment:



Class Diagram

State Chart

